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# Age of the Alleghenian folding in the Central Appalachians

Rob Van der Voo
Department of Geology and Mineralogy
University of Michigan
Ann Arbor, Michigan 48109

## ABSTRACT

Paleomagnetic studies of folded Ordovician and Silurian red beds from the Valley and Ridge province of Virginia to Pennsylvania reveal secondary magnetizations that postdate the folding. The pole position for these secondary magnetizations is statistically different from all North American poles younger than Early Permian. Further comparison with the apparent polar wander path of cratonic North America implies that the Alleghenian folding in the Central Appalachians is at least as old as Early Permian and is probably of Late Carboniferous age.

## INTRODUCTION

In the Valley and Ridge province of central Pennsylvania, western Maryland, Virginia, and eastern West Virginia, the youngest formations clearly affected by Alleghenian folding are of Pennsylvanian age. Farther west in West Virginia, the Dunkard Group is thought to be of Late Carboniferous–earliest Permian age; it conformably overlies the Upper Carboniferous Monongahela Formation (Colton, 1970). Because younger Permian and Lower Triassic formations are lacking, there are few constraints on the minimum age of the Alleghenian deformation in this area (see, for example, Palmer, 1978).

Perry (1978) has argued that Alleghenian deformation in Pennsylvania started in Early Carboniferous time, as he observed Lower Mississippian debris-flow deposits containing pebbly mudstones and clasts of Middle and Upper Devonian rocks from western Maryland and Pennsylvania. This, however, does not provide information about either the peak or the end of the folding phase(s).

There is, on the other hand, an interesting paleomagnetic constraint for the age of the folding. It is based on the occurrence of secondary magnetizations that can be shown to be of postfolding age. The pole position derived from the directions of magnetization falls near the Late Carboniferous paleomagnetic poles for cratonic North America and is significantly different from all poles younger than Early Permian. In the next section, the details and statistics of the paleomagnetic argument will be examined.

## THE POSTFOLDING MAGNETIZATIONS

Three studies (French and Van der Voo, 1977; Van der Voo and French, 1977; French, 1978) have discussed the paleomagnetism of the Upper Ordovician Juniata Formation and the Middle Silurian Rose Hill Formation from the Central Appalachians (Fig. 1). Demagnetization with thermal, chemical, and alternating field techniques yielded presumably original magnetizations of prefolding age, with normal and reversed polarities and moderate to steep inclinations; the mean normal-polarity directions of magnetization were declination ($D$) = 348.6° and inclination ($I$) = −31.8° for the Juniata (Van der Voo and French, 1977) and $D$ = 330.0° and $I$ = −43.7° for Rose Hill siderite-bearing beds, as well as red and gray sandstones (French, 1978).

In all three studies, most red sandstone and siltstone samples were observed to have secondary, lower-blocking-temperature directions of magnetization that did not pass the fold test of Graham (1949). This implies that the secondary magnetizations are of postfolding age. Site-mean directions have been calculated for those sites that have more than two samples with unambiguous secondary directions of magnetization (Table 1, Fig. 2). In Table 1, their overall mean pole is also listed with the associated statistical parameters, $k_1$ and $k_2$, before and after correction for the tilt of the strata, respectively (Fisher, 1953). It can be seen that the ratio $k_1/k_2$ = 5.4; this indicates that the failure of the fold test is significant at the 99% confidence level (McElhinny, 1964).

## THE SECONDARY PALEOMAGNETIC POLE

Because of the widespread geographical distribution of the sampling sites, a mean pole position has been calculated from the site-mean poles. This pole is located at

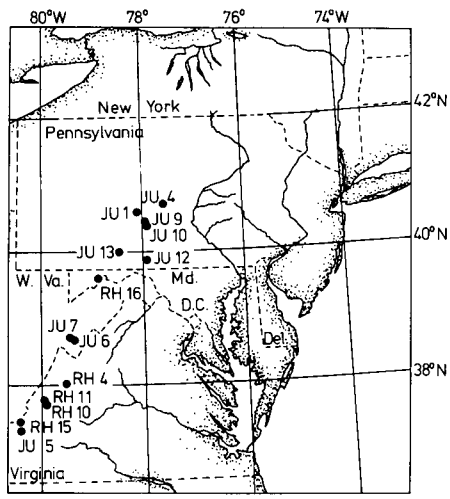



Figure 1. Map of the Central Appalachians showing the site locations of the Rose Hill (RH) and Juniata (JU) Formations where unambiguous secondary magnetizations have been observed.

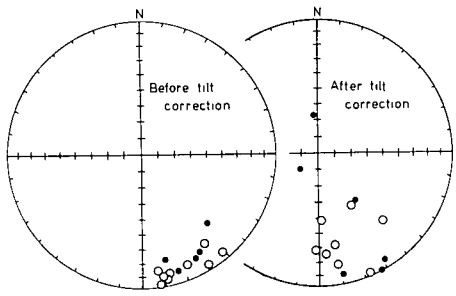



Figure 2. Equal-area projections of site-mean directions of secondary magnetization, before and after correction for the tilt of the strata. Open circles represent projections on the upper hemisphere; solid circles represent projections on the lower hemisphere.

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TABLE 1. SECONDARY MAGNETIZATIONS AND POLE POSITIONS, JUNIATA AND ROSE HILL FORMATIONS, VALLEY AND RIDGE PROVINCE, CENTRAL APPALACHIANS

| Site | No.* | $D$† (°) | $I$† (°) | Pole position† lat (°N) | Pole position† long (°E) | $D$§ (°) | $I$§ (°) | $k$# | $\alpha_{95}$# | Pole position§ lat (°N) | Pole position§ long (°E) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| JU 1 | 5 | 149.8 | +16.9 | 33.5 | 138.4 | 148.9 | +5.7 | 30.2 | 14.2 | 38.1 | 142.8 |
| JU 4 | 4 | 169.0 | -5.7 | 51.0 | 120.0 | 178.4 | -48.8 | 4.5 | 48.8 | 79.1 | 109.8 |
| JU 5 | 5 | 168.4 | +21.0 | 40.6 | 114.6 | 171.2 | -31.4 | 9.2 | 26.7 | 68.3 | 122.8 |
| JU 6 | 8 | 157.8 | -13.3 | 52.3 | 138.4 | 157.7 | -0.4 | 15.3 | 14.6 | 46.4 | 134.0 |
| JU 7 | 9 | 153.4 | +14.3 | 37.8 | 134.8 | 152.2 | +0.1 | 21.4 | 11.4 | 43.6 | 140.7 |
| JU 9 | 3 | 171.0 | -1.7 | 49.6 | 116.1 | 177.7 | -25.8 | 56.3 | 16.5 | 63.1 | 107.0 |
| JU 10 | 3 | 173.5 | -14.1 | 56.3 | 113.8 | 182.0 | -28.1 | 6.7 | 52.1 | 64.5 | 97.6 |
| JU 12 | 4 | 167.5 | -6.7 | 51.8 | 122.5 | 171.0 | -17.2 | 24.0 | 19.1 | 57.9 | 118.9 |
| JU 13 | 5 | 149.1 | -5.4 | 43.4 | 146.4 | 230.9 | +75.6 | 3.2 | 50.5 | 20.6 | 259.3 |
| RH 4 | 6 | 145.4 | -20.0 | 48.5 | 157.8 | 149.2 | -52.5 | 7.3 | 26.5 | 64.6 | 188.7 |
| RH 10 | 6 | 137.3 | +29.7 | 23.0 | 145.2 | 137.3 | -31.3 | 15.5 | 17.6 | 47.2 | 173.0 |
| RH 11 | 3 | 141.7 | -6.0 | 40.7 | 154.8 | 352.2 | +65.7 | 17.4 | 30.4 | 78.3 | 253.3 |
| RH 15 | 6 | 163.0 | +10.0 | 44.7 | 123.8 | 144.0 | +53.0 | 41.0 | 11.0 | 11.5 | 129.6 |
| RH 16 | 43 | 169.1 | -5.0 | 51.6 | 118.9 | 169.4 | +8.4 | 31.0 | 3.9 | 45.1 | 116.3 |
| Mean (14 sites) | | 158.5 | +0.9 | 45.6 | 132.7 | 162.9 | -5.1 | | | 54.8 | 128.6 |
| *Statistical parameters*# | | $k$ = 18.5 | | $k_1$ = 34.2 | | $k$ = 3.0 | | | | $k_2$ = 6.3 | |
| | | $\alpha_{95}$ = 9.5° | | $A_{95}$ = 6.9° | | $\alpha_{95}$ = 27.7° | | | | $A_{95}$ = 17.2° | |

*Note*: Sites JU 1 through JU 13 are described in Van der Voo and French (1977). Sites RH 4 through RH 15 from French and Van der Voo (1977) with site coordinates lat 38.008°N, long 79.535°W (site 4); lat 37.713°N, long 79.907°W (site 10); lat 37.740°N, long 79.910°W (site 11); lat 37.388°N, long 80.393°W (site 15). Site RH 16 from French (1978), with coordinates lat 39.560°N, long 78.820°W.

*Number of samples or sites used in the statistical analysis.

†Before correction for the tilt of the strata.

§After correction for the tilt of the strata.

#Fisher (1953); $\alpha_{95}$ is used for mean directions, $A_{95}$ for mean poles; the means are listed for each column of directions or poles.

TABLE 2. PERMIAN POLE POSITIONS FOR NORTH AMERICA

| Pole no.* | Pole position lat (°N) | Pole position long (°E) | Source |
|---|---|---|---|
| | | | *Lower Permian* |
| 7.73 | 44.0 | 122.0 | Dunkard, W.Va. |
| 7.212 | 43.6 | 119.6 | Elephant Canyon, Utah |
| 7.177 | 42.7 | 121.9 | Cutler, Colo. |
| 7.211 | 44.1 | 115.4 | Cutler, Utah |
| 7.147 | 31.5 | 128.7 | Lower Maroon, Colo. |
| 7.160 | 41.2 | 117.7 | Wolfcampian, SW U.S. |
| 7.161 | 44.8 | 114.6 | Leonardian, SW U.S. |
| Mean | 41.8 | 120.2 | |
| $k$ = 186.6; $A_{95}$ = 4.43°† | | | |
| | | | *Upper Permian* |
| 7.117 | 51.7 | 112.8 | Prince Edward I. sill |
| 7.95 | 47.0 | 103.0 | Toroweap, Ariz. |
| 7.148 | 59.8 | 102.4 | Upper Maroon, Colo. |
| 7.162 | 51.1 | 125.4 | Guadalupian, SW U.S. |
| 7.163 | 54.9 | 118.9 | Ochoan, SW U.S. |
| Mean | 53.2 | 112.6 | |
| $k$ = 111.6; $A_{95}$ = 7.27°† | | | |

*From Irving and others (1976); other poles in that list are excluded for the following reasons: combination poles (7.205, 7.215, 7.206); no demagnetization (7.47 through 7.52); imprecise age (7.149); not enough samples (7.96); results superseded by later poles (7.44, 7.45, 7.97).

†Statistical parameters (Fisher, 1953).

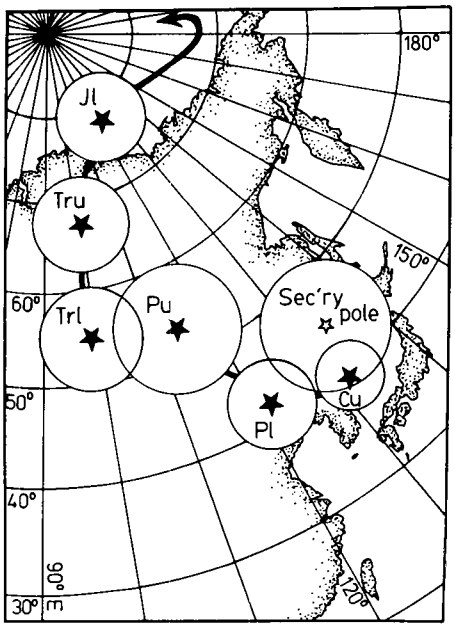



**Figure 3. The pole obtained from the secondary magnetizations (open star) and the relevant segment of the apparent polar wander path (Van der Voo and French, 1974) for cratonic North America with mean poles (solid stars). The cones of 95% confidence ($A_{95}$) have been drawn as circles around the mean poles. Abbreviations and sources are as follows: Cu = Late Carboniferous (Van der Voo and French, 1974); Pl = Early Permian and Pu = Late Permian (from Table 2); Trl = Early Triassic and Tru = Late Triassic, and Jl = Early Jurassic (all from Van der Voo and French, 1974).**

lat 45.6°N, long 132.7°E and has associated statistical parameters of $k$ = 34.2 and $A_{95}$ = 6.9°. In Figure 3, this mean secondary pole is compared with the apparent polar wander path (Van der Voo and French, 1974) for the North American craton. It can be seen at a glance that the secondary pole is quite different from all poles younger than Permian. Upon more detailed examination, Figure 3 shows that the cone of confidence of the secondary pole overlaps only with the cone of confidence of the Late Carboniferous pole and (barely) with that of the Early Permian pole (Table 2). The secondary pole is significantly different at the 95% confidence level from the Late Permian pole. The secondary magnetizations are therefore taken to be earliest Permian or older. This implies that the Alleghenian folding is at least of Early Permian and, presumably, of Late Carboniferous age.

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